1 The Deutsch-Jozsa Algorithm

1.1 The Problem

Suppose you are given a function f that takes n bits as input and produces a single bit as output:

$$f : \{0, 1\}^n \to \{0, 1\}$$

You do not know what the function does internally, but you are told it is *guaranteed* to be one of two types:

- **Constant** — the function returns the same output (either always 0 or always 1) for every possible input.
- **Balanced** — the function returns 0 for exactly half of all possible inputs, and 1 for the other half.

Your task is to determine which type it is.

Classical Approach

On a classical computer, you would have to evaluate f on different inputs one at a time. In the *worst case*, you could get the same output for the first $\frac{2^n}{2}$ queries and still not be certain whether the function is constant or balanced — a single additional query is needed to make that determination. This gives a worst-case query complexity of:

$$\frac{2^n}{2} + 1$$

For large n, this grows exponentially.

Quantum Approach

Using the **Deutsch-Jozsa algorithm**, a quantum computer can solve this problem with only a *single* query to f, regardless of n.

1.2 The Quantum Oracle

To use a function f on a quantum computer, it must be implemented as a **quantum oracle**. A quantum oracle U_f is a unitary operation that encodes f in a way that is reversible — a fundamental requirement for quantum computation.

The oracle takes two registers as input:

- An **input register** of n qubits, holding the state $|x\rangle$.
- A **target qubit**, holding the state $|y\rangle$.

It acts on them as follows:

$$U_f\, |x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

where $\oplus$ denotes XOR. The input register is left unchanged, and the target qubit is flipped if and only if $f(x) = 1$.

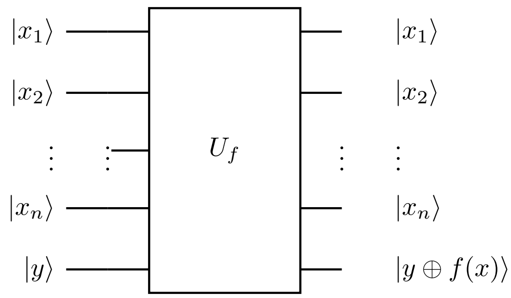

Figure 1: Quantum circuit for the oracle U_f. The top n qubits form the input register and are returned unchanged. The bottom target qubit is returned as $|y \oplus f(x)\rangle$.

1.3 The Phase Oracle

In the Deutsch-Jozsa algorithm, the target qubit is initialized to the state $|-\rangle$, defined as:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

When the oracle is applied with the target qubit in this state, something special happens. Let us see what U_f does to $|x\rangle|-\rangle$:

$$U_f\,|x\rangle|-\rangle = U_f\,|x\rangle \cdot \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\big(U_f\,|x\rangle|0\rangle - U_f\,|x\rangle|1\rangle\big) = |x\rangle \cdot \frac{|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}}$$

- If $f(x) = 0$: the target qubit is unchanged, giving $\dfrac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$.
- If $f(x) = 1$: the two terms swap, giving $\dfrac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle$.

In both cases, the target qubit remains $|-\rangle$ (up to a global phase), and the effect of $f(x)$ is instead imprinted as a *phase* on the input register:

$$\boxed{U_f\,|x\rangle|-\rangle = (-1)^{f(x)}\,|x\rangle|-\rangle}$$

Since the target qubit is unaffected, it can be ignored from this point on. The oracle has effectively become a **phase oracle** — one that encodes $f(x)$ as a phase factor $(-1)^{f(x)}$ on the input state $|x\rangle$.

1.4 The Algorithm

We now have everything we need. The full Deutsch-Jozsa circuit is shown below.

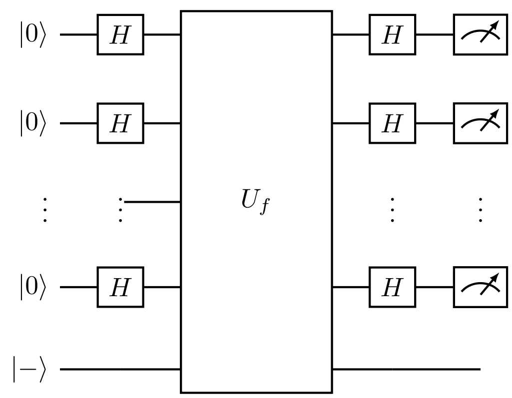

Figure 2: The Deutsch-Jozsa circuit. The top n qubits are the input register and the bottom qubit is the target, pre-initialized to $|-\rangle$.

Step 1: Initialize

The input register is set to $|0\rangle^{\otimes n}$ and the target qubit is pre-initialized to $|-\rangle$. The full system state is:

$$|\psi_0\rangle = |0\rangle^{\otimes n}|-\rangle$$

Step 2: Apply Hadamard to the input register

A Hadamard gate is applied to each of the n input qubits. The Hadamard gate puts a single qubit into an equal superposition:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Applying it to all n qubits simultaneously creates an equal superposition of *all* 2^n possible input states:

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle$$

The system state is now:

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle\,|-\rangle$$

Step 3: Apply the Oracle

The phase oracle U_f is applied. As derived earlier, it maps $|x\rangle|-\rangle \mapsto (-1)^{f(x)}|x\rangle|-\rangle$, so:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}|x\rangle\,|-\rangle$$

The target qubit $|-\rangle$ is unaffected and can be dropped from further analysis. We focus on the input register:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}|x\rangle$$

Step 4: Apply Hadamard again to the input register

A second round of Hadamard gates is applied to the input register. To see what this does, we first need to understand how H acts on a single qubit $|x_i\rangle$ where $x_i \in \{0,1\}$.

We already know:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Both cases can be written in a single compact form. Notice that the sign on $|1\rangle$ is $+1$ when $x_i = 0$ and -1 when $x_i = 1$, which is exactly $(-1)^{x_i}$. The sign on $|0\rangle$ is always $+1$, which is $(-1)^0$. So we can write:

$$H|x_i\rangle = \frac{1}{\sqrt{2}} \sum_{z_i\in\{0,1\}} (-1)^{x_i z_i}|z_i\rangle$$

You can verify: when $x_i = 0$, all exponents are 0 so both terms are $+1$; when $x_i = 1$, the $z_i = 1$ term picks up a $(-1)^1 = -1$ sign.

Now $H^{\otimes n}$ applies H independently to each of the n qubits. Since the qubits are independent, the result is the tensor product of each single-qubit Hadamard:

$$H^{\otimes n}|x\rangle = H|x_0\rangle \otimes H|x_1\rangle \otimes \cdots \otimes H|x_{n-1}\rangle$$

$$= \left(\frac{1}{\sqrt{2}}\sum_{z_0}(-1)^{x_0 z_0}|z_0\rangle\right)\otimes\left(\frac{1}{\sqrt{2}}\sum_{z_1}(-1)^{x_1 z_1}|z_1\rangle\right)\otimes\cdots\otimes\left(\frac{1}{\sqrt{2}}\sum_{z_{n-1}}(-1)^{x_{n-1} z_{n-1}}|z_{n-1}\rangle\right)$$

Expanding this tensor product collapses all the individual sums into a single sum over all n-bit strings z. The $\frac{1}{\sqrt{2}}$ factors combine to give $\frac{1}{\sqrt{2^n}}$, and the phases multiply together:

$$(-1)^{x_0 z_0} \cdot (-1)^{x_1 z_1} \cdots (-1)^{x_{n-1} z_{n-1}} = (-1)^{x_0 z_0 + x_1 z_1 + \cdots + x_{n-1} z_{n-1}}$$

This exponent is the **bitwise inner product** of x and z, written as $x \cdot z$:

$$x \cdot z = x_0 z_0 \oplus x_1 z_1 \oplus \cdots \oplus x_{n-1} z_{n-1}$$

So the full n-qubit Hadamard transform is:

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z\in\{0,1\}^n} (-1)^{x\cdot z}|z\rangle$$

Applying this to $|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle$:

$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} \cdot \frac{1}{\sqrt{2^n}} \sum_z (-1)^{x \cdot z} |z\rangle = \sum_{z \in \{0,1\}^n} \left(\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x) + x \cdot z} \right) |z\rangle$$

Step 5: Measure

We measure the input register. The amplitude of the all-zeros state $|0\rangle^{\otimes n}$ is obtained by setting $z = 0 \cdots 0$, which makes $x \cdot z = 0$ for all x:

$$\text{amplitude of } |0\rangle^{\otimes n} = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)}$$

1.5 Interpreting the Result

The key is to look at what this amplitude evaluates to in each case.

If f is constant: $f(x)$ is the same value for all x, so $(-1)^{f(x)}$ is either $+1$ for all x or -1 for all x. Either way, all 2^n terms in the sum are identical and the sum does not cancel:

$$\frac{1}{2^n} \sum_x (-1)^{f(x)} = \frac{1}{2^n} \cdot 2^n \cdot (\pm 1) = \pm 1$$

The amplitude of $|0\rangle^{\otimes n}$ has magnitude 1, meaning the measurement outcome is $|0\rangle^{\otimes n}$ *with certainty.*

If f is balanced: $f(x) = 0$ for exactly half the inputs and $f(x) = 1$ for the other half. The $+1$ and -1 terms cancel out exactly:

$$\frac{1}{2^n} \sum_x (-1)^{f(x)} = \frac{1}{2^n} \left(2^{n-1} \cdot (+1) + 2^{n-1} \cdot (-1) \right) = 0$$

The amplitude of $|0\rangle^{\otimes n}$ is 0, so the measurement outcome is *never* $|0\rangle^{\otimes n}$.

The conclusion is simple:

- **All measurement bits are** $0 \Rightarrow f$ is **constant**.
- **Any measurement bit is** $1 \Rightarrow f$ is **balanced**.

This is determined with a **single query** to the oracle, regardless of how many input bits n there are.

1.6 The Deutsch Algorithm

The Deutsch algorithm is simply a special case of the Deutsch-Jozsa algorithm where the number of inputs is $n = 1$. Everything works exactly the same way, just with a single input qubit instead of n.

To get comfortable with how the algorithm works, it is highly recommended that you work through the Deutsch algorithm from scratch on your own — follow each step, do the math, and see if you can arrive at the conclusion of whether the function is constant or balanced yourself.